Databases - Task I

Topic 15. - Planes, producers, airlines

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1. Topic analysis

Client

- A medium-sized airline company like LOT Polish Airlines.

Goals of the system

- Provides a centralized source of information to enhance collaboration across different departments of the company.
- Facilitates managing the airplanes' maintenance and inspection events.
- Simplifies generating analytical reports covering inspection and maintenance history, inventory usage, etc.

Users and sample usage scenarios

User	**Scenarios**
Inspection Specialist	- Record inspections of airplanes. - Initiate maintenance tasks for failed inspections. - Update airplane status based on inspection results.
Maintenance Coordinator	- Manage maintenance events initiated via inspections. - Retrieve the maintenance history of airplanes. - Record inventory usage during maintenance.
Operations Analyst	- Analyze airplane repair frequency. - Monitor inventory usage and reorder levels. - Generate reports on inspection outcomes.

Assumptions

- Inspections and maintenance tasks follow a linear and repeatable process:
    - An initiating inspection confirms quality or identifies issues,
    - Maintenance is performed,
    - A validating inspection confirms resolution or triggers further work.
- Employees have clearly defined roles (Inspection Specialists, Maintenance Coordinators) with proper authority.
- The system preserves both inspection and maintenance history to ensure regulatory compliance.
- Locations (airports, workshops) are sufficiently captured and maintained within the database to track the physical availability of airplanes and parts.

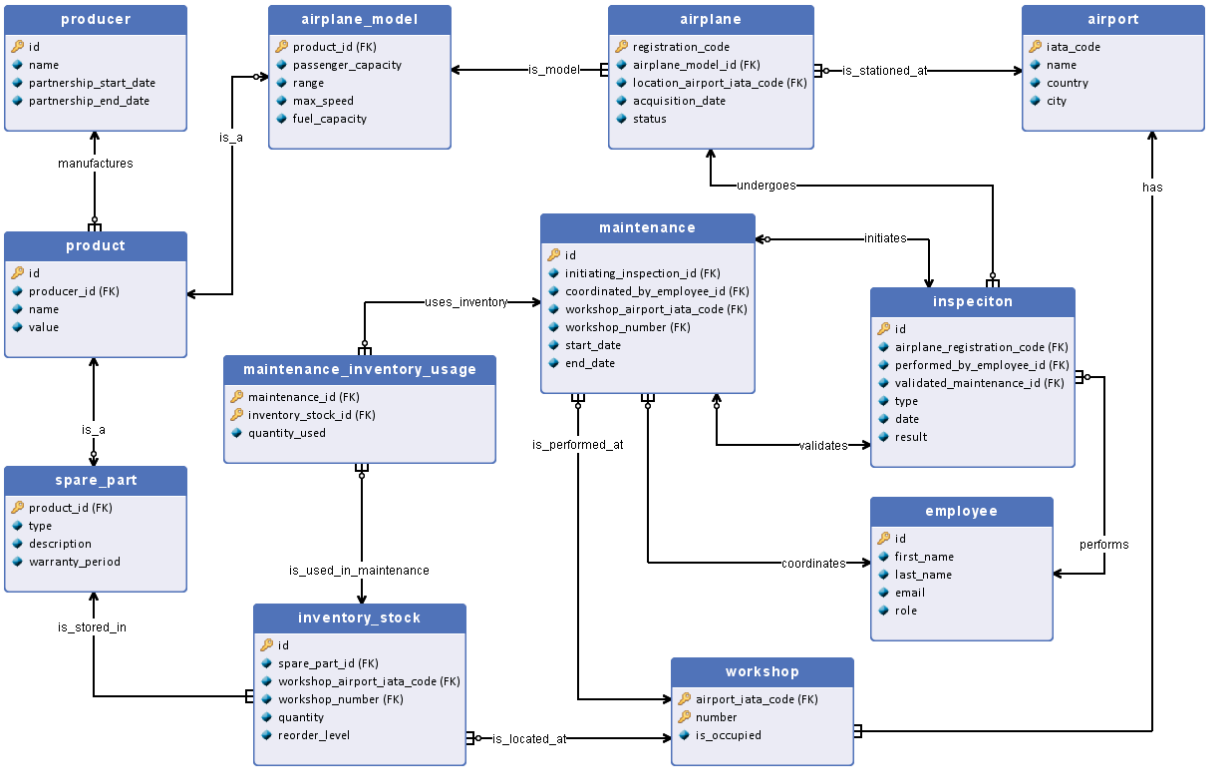
Limitations

- The design assumes that maintenance and inspections are distinct entities and does not explicitly handle concurrent or overlapping tasks on the same airplane.
- Inspections and maintenance tasks are tied to individual employees. Multi-party workflows are not supported by the system.
- The scope of the system excludes managing flights, passenger logistics, and detailed transactions (deliveries, leases, etc.) between the airline, aircraft producers, and airports.

Example queries in natural language

- List all maintenance events performed on an airplane in the last year.
- Check the result of the most recent inspection of an airplane.
- List currently available workshops at a specific airport.
- Calculate the number of inspections that initiated maintenance in the last year.
- List all spare parts used during the last maintenance of an airplane, including quantity.
- Check the quantity of a particular spare part available at a specific workshop.
- Calculate the total value of spare parts used in the last month.
- For the last year, calculate the average number of days it takes to finish maintenance.

2. Entity-relationship diagram



3. Description of ERD

Entity sets

**In accordance with the system's assumptions, once created, entities are stored permanently to ensure regulatory compliance (audit events require full aircraft lifecycle history).*

- **producer, ~5 entries**
  Companies that manufacture airplanes and spare parts used in the airline's operations. Added when the airline procures from a new producer.
  - *id* (PK) - INT, autoincrement
  - *name* - producer company name, VARCHAR(30), starting with a capital letter
  - *partnership_start_date* - starting date of the partnership with the producer, DATE, YYYY-MM-DD format, ranging from 1970-01-01 to present date + 1 year
  - *partnership_end_date* - ending date of the partnership with the producer, DATE, YYYY-MM-DD format, ranging from the present date + 1 year, or NULL for indefinite contracts

- **product, ~530 entries (airplane_model + spare_part)**
  Base entity representing all airplanes and spare parts. Added as a result of registering a new airplane model or spare part.
    - *id* (PK) - INT, autoincrement
    - *producer_id* (FK)
    - *name* - name of the product, VARCHAR(30), starting with a capital letter
    - *value* - cost in USD rounded up to hundreds, INT, only positive values

- **airplane_model, ~30 entries**
  Distinct airplane models manufactured by producers. Added when the airline purchases a new model.
    - *product_id* (PK + FK)
    - *passenger_capacity* - number of passengers that can be boarded, INT, only positive values
    - *range* - maximum number of kilometers per one flight rounded down to hundreds, INT, only positive values
    - *max_speed* - maximum speed in km/h achieved by the airplane rounded down to tens, INT, only positive values
    - *fuel_capacity* - tank capacity in liters rounded down to tens, INT, only positive values

- **spare_part, ~500 entries**
  Spare parts used in maintenance. Added when a new spare part is acquired.
    - *product_id* (PK + FK)
    - *type* - category of the component, VARCHAR(10), fixed possible values:
      - *electrical* (e.g. navigation light)
      - *mechanical* (e.g. landing gear)
      - *hydraulic* (e.g. control valve)
      - *avionics* (e.g. radar antenna)
      - *structural* (e.g. wing rib)
      - *cabin* (e.g. passenger seat)
      - *safety* (e.g. fire extinguisher)
    - *description* - technical details of the spare part, VARCHAR(500)
    - *warranty_period* - warranty period in months, INT, only positive values

- **airplane, ~100 entries**
  Individual airplanes operated by the airline. Added upon acquisition.
    - *registration_code* (PK) - unique identifier assigned when the airplane is registered in the airline's fleet, CHAR(6), format SP-L** where ** is a unique combination of two uppercase letters
    - *airplane_model_id* (FK)
    - *location_airport_iata_code* (FK)
    - *acquisition_date* - date since the airplane belongs to the airline's fleet, DATE, YYYY-MM-DD format, ranging from 1970-01-01 to the present date
    - *status* - indicates current airplane availability, VARCHAR(12), fixed possible values:
      - *active* - fully operational
      - *inspection* - under inspection
      - *maintenance* - under maintenance
      - *suspended* - temporarily withdrawn from service
      - *retired* - permanently removed from service

- **airport, ~50 entries**
  Airports involved in airline operations. Added as new operating airports are established.
    - *iata_code* (PK) - 3-letter International Air Transport Association code identifying the airport, CHAR(3), only uppercase letters
    - *name* - full name of the airport, VARCHAR(60), only letters and spaces, starting with a capital letter,
    - *country* - country of location, VARCHAR(50), only letters and spaces, starting with a capital letter
    - *city* - city of location, VARCHAR(60), only letters and spaces, starting with a capital letter
- **workshop, ~200 entries**
  Specific facilities at airports where maintenance tasks are performed. Added when a new airport is added.
    - *airport_iata_code* (PK + FK)
    - *number* (PK) - workshop number unique on a given airport, INT, only positive values
    - *is_occupied* - indicates whether the workshop is currently occupied by an airplane, 0/1 (BIT in MS SQL Server)

- **inspection, ~10000 entries/year**
  Records of inspections performed on airplanes. Added when an airplane must be inspected.
    - *id* (PK) - INT, autoincrement
    - *airplane_registration_code* (FK)
    - *performed_by_employee_id* (FK)
    - *validated_maintenance_id* (FK)
    - *type* - indicates the type of the inspection, VARCHAR(16), fixed possible values:
      - *pre-flight* - performed before flights
      - *post-flight* - performed after landing
      - *routine* - scheduled inspections
      - *post-maintenance* - verification after maintenance
    - *date* - date of the inspection, DATE in YYYY-MM-DD format, ranging from 1970-01-01 to present date + 1 year
    - *result* - indicates the inspection status/outcome, VARCHAR(10), fixed possible values:
      - *scheduled*
      - *pending*
      - *passed*
      - *failed*

- **maintenance, ~5000 entries/year**
  Records of maintenance events performed on airplanes. Added as a response to a failed inspection.
    - *id* (PK) - INT, autoincrement
    - *initiating_inspection_id* (FK)
    - *coordinated_by_employee_id* (FK)
    - *workshop_airport_iata_code* (FK)
    - *workshop_number* (FK)
    - *start_date* - starting date of the maintenance task, DATE, YYYY-MM-DD format, ranging from 1970-01-01 to present date + 1 year, or NULL if the date is yet to be determined

    - *end_date* - ending date of the maintenance task, DATE, YYYY-MM-DD format, ranging from 1970-01-01 to present date + 1 year, or NULL if the maintenance has not started yet/is still in progress

- **inventory_stock, ~1000 entires**
  All spare parts and their quantities stored at different workshops.
    - *id* (PK) - INT, autoincrement
    - *spare_part_id* (FK)
    - *workshop_airport_iata_code* (FK)
    - *workshop_number* (FK)
    - *quantity* - number of the given part stored in a particular workshop, INT, only non-negative values
    - *reorder_level* - the minimum quantity of the spare part that should be stored (quantity dropping below this threshold requires restocking), INT, only positive values

- **maintenance_inventory_usage, ~10000 entries/year**
  Links maintenance events to inventory stock used, tracking quantities. Added when maintenance consumes inventory.
    - *maintenance_id* (PK + FK)
    - *inventory_stock_id* (PK + FK)
    - *quantity_used* - quantity of a spare part used from a particular inventory stock, INT, only positive values

- **employee, ~100 entries**
  Personnel working for the airline. Added when an employee joins as either Inspection Specialist or Maintenance Coordinator.
    - *id* (PK) - INT, autoincrement
    - *first_name* - VARCHAR(30), only letters, starting with a capital letter
    - *last_name* - VARCHAR(30), only letters, starting with a capital letter
    - *email* - internal email address of an employee, VARCHAR(40), format #.***@lot.pl where # is the first letter of the employee's first name and *** is the employee's last name. In case of multiple employees with the same email pattern, the first one receives it and others get an additional digit before the dot indicating the order of joining the company.
    - *role* - specifies the responsibilities of the employee, VARCHAR(24), fixed possible values:
      - *Inspection Specialist* - responsible for creating and updating *inspection* entities and initiating *maintenance* if needed
      - *Maintenance Coordinator* - responsible for updating *maintenance* and creating *maintenance_inventory_usage* entities when needed

Relationships

- **manufactures (producer 1:0..n product)**
  Represents which producers manufacture which products. Each product is manufactured by exactly one producer, but producers may manufacture multiple products. Optionality is included as it is possible to add a producer with future partnership start date without registering any products.

- **is_a (airplane_model 0..1:1 product)**
  Defines airplane_model as a subset of product. Each airplane model is also a product and must have its base properties defined in the corresponding product entity. A product can also be a spare part, thus it can exist without an airplane model.

- **is_a (spare_part 0..1:1 product)**
  Defines spare_part as a subset of product. Each spare part is also a product and must have its base properties defined in the corresponding product entity. A product can also be an airplane model, thus it can exist without a spare part.

- **is_model (airplane 1..n:1 airplane_model)**
  Represents which airplanes are of which model. Each airplane is of exactly one model, but the company may utilize multiple airplanes of the same model. Optionality is excluded as the database stores only the models that were/are used by the airline.

- **is_stationed_at (airplane 0..n:1 airport)**
  Represents which airplanes are stationed at specific airports. Each airport can hold multiple airplanes at once, but may also be empty.

- **has (airport 1:1..n workshop)**
  Represents which workshops belong to which airports. Each airport has at least one workshop that can be used for maintenance tasks.

- **undergoes (airplane 1:0..n inspection)**
  Represents which airplane goes through particular inspection tasks. Each inspection is performed on an individual airplane. An airplane may undergo multiple inspection tasks during its lifecycle. Optionality is included as a newly acquired airplane did not go through any inspections.

- **initiates (inspection 1:0..1 maintenance)**
  Represents which inspection triggered which maintenance task. Zero or one maintenance is initiated directly by each inspection, as successful inspections do not require further maintenance. Each maintenance is performed as a response to an individual inspection.

- **validates (inspection 1:0..1 maintenance)**
  Represents which inspection is validating post-maintenance quality. Each inspection may validate only one maintenance task, but it may also be performed as a routine task without prior maintenance.

- **is_performed_at (maintenance 0..n:1 workshop)**
  Represents which workshop is used for the purpose of performing particular maintenance tasks. Each maintenance is performed in exactly one workshop. Each workshop can be used for one maintenance at a time, but may include zero or more maintenance tasks in its history.

- **uses_inventory (maintenance 1:0..n maintenance_inventory_usage)**
  Used for the many-to-many relationship between maintenance and inventory_stock. Each maintenance may use multiple inventory stocks (spare parts available in particular workshops), but may also be completed without using any spare parts.

- **is_used_in_maintenance (inventory_stock 1:0..n maintenance_inventory_usage)**
  Used for the many-to-many relationship between maintenance and inventory_stock. Each inventory stock may be used in multiple maintenance events, but there may be stocks that were never used.

- **is_stored_in (spare_part 1:1..n inventory_stock)**
  Represents which spare parts are stored in which inventory stocks (tracking quantity and assigned to specific workshop locations). The same spare parts can be stored in multiple inventory stocks, but an inventory stock is defined for one part only (to track its availability). Optionality is excluded as new spare parts are recorded on acquisition and they must always be stored somewhere.

- **is_located_at (inventory_stock 0..n:1 workshop)**
  Represents the workshop locations of particular inventory stocks. Each workshop may hold multiple inventory stocks (also with quantity=0 if a spare part was stored and used), but may also be empty if nothing was ever stored in it.

- **performs (employee 1:0..n inspection)**
  Represents who performs which inspections. Each employee may have conducted multiple inspections during their work experience within the company, but may also have zero if they are not an Inspection Specialist or in case of a new employee.

- **coordinates (employee 1:0..n maintenance)**
  Represents who coordinates which maintenance tasks. Each employee may have supervised many maintenance tasks during their work experience within the company, but may also have zero if they are not a Maintenance Coordinator or in case of a new employee.

4. Relational database schema

- producer (id, name, partnership_start_date, partnership_end_date)

- product (id, producer_id REF producer, name, value)

- airplane_model (product_id REF product, passenger_capacity, range, max_speed, fuel_capacity)

- spare_part (product_id REF product, type, description, warranty_period)

- airplane (registration_code, airplane_model_id REF airplane_model, location_airport_iata_code REF airport, acquisition_date, status)

- airport (iata_code, name, country, city)

- workshop (aiport_iata_code REF airport, number, is_occupied)

- inspection (id, airplane_registration_code REF airplane, performed_by_employee_id REF employee, validated_maintenance_id REF maintenance, type, date, result)

- maintenance (id, initiating_inspection_id REF inspection, coordinated_by_employee_id REF employee, (workshop_airport_iata_code, workshop_number) REF workshop, start_date, end_date)

- inventory_stock (id, spare_part_id REF spare_part, (workshop_airport_iata_code, workshop_number) REF workshop, quantity, reorder_level)

- maintenance_inventory_usage (maintenance_id REF maintenance, inventory_stock_id REF inventory_stock, quantity_used)

- employee (id, first_name, last_name, email, role)